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MULTIPLE CHOICE QUESTIONS

Sem: I	**Branch: CSE/ISE/CCE/AIM/AI&DS/R&AI/CYB/CBS**
Course code: MAT103	**Course Title: CALCULUS AND DIFFERENTIAL EQUATIONS**

MAT103

UNIT-I- Differential Calculus

1.	If φ be the angle between the tangent and radius vector at any point on the curve, $r = f(\theta)$ then $\tan\varphi$ equals to:	
	a) $\frac{dr}{ds}$	b) $r\frac{d\theta}{ds}$
	c) $r\frac{d\theta}{dr}$	d) $\frac{d\theta}{dr}$
2.	Curvature of a straight line is :	
	a) ∞	b) 0
	c) 1	d) none of these
3.	The angle between the radius vector $r = a\sin\theta$ and tangent to the curve at any point is $\varphi =$	
	a) $\frac{\theta}{2}$	b) θ
	c) 0	d) $\frac{\pi}{2}$
4.	The radius of curvature of the curve $y = x^2$ when $x = 0$ is :	
	a) $\frac{1}{\sqrt{2}}$	b) $\sqrt{2}$
	c) 2	d) $\frac{1}{2}$
5.	The polar form of the cartesian equation $x^2 + y^2 = 9$ is :	
	a) $r^2 = 81$	b) $r^2 = 18$
	c) $r = 9$	d) $r = 3$
6.	The radius of curvature of the curve $y = e^x$ at the point $(0, 1)$ is:	
	a) $2\sqrt{2}$	b) $\sqrt{2}$
	c) 2	d) $\frac{\sqrt{2}}{2}$
7.	For the polar curve $r = f(\theta)$, the relation between θ and coordinates (x, y) is:	

	a) $tan\theta = \frac{x}{y}$	b) $tan\theta = \frac{y}{x}$
	c) $x = rcos\theta$	d) $x = rsin\theta$
8.	The radius of curvature for the curve $y = f(x)$ is $\rho =$	
	a) $\frac{(1+{y_2}^2)^{3/2}}{y_1}$	b) $\frac{(1+{y_1}^2)^{3/2}}{y_2}$
	c) $\frac{(1+{y_1}^2)^{2/3}}{y_2}$	d) $\frac{(1-{y_1}^2)^{3/2}}{y_2}$
9.	The Maclaurin's series expansion of the function $f(x) = e^x$ is:	
	a) $1 - x - \frac{x^2}{2} - \frac{x^3}{6} + \cdots$	b) $1 - x + \frac{x^2}{2} - \frac{x^3}{6} + \cdots$
	c) $1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots$	d) $-1 + x - \frac{x^2}{2} + \frac{x^3}{6} - \cdots$
10.	The radius of curvature at any point of the circle $x^2 + y^2 = 16$ is:	
	a) 4	b) 12
	c) 16	d) 6
11.	If φ be the angle between the tangent and radius vector at any point on the curve, $r = f(\theta)$ then $\cot\varphi$ is equals to:	
	a) $r\frac{dr}{d\theta}$	b) $\frac{1}{r}\frac{dr}{d\theta}$
	c) $r\frac{d\theta}{dr}$	d) $\frac{1}{r}\frac{d\theta}{dr}$
12.	The Taylor's series expansion of the function $f(x)$ about the point $x = a$ is:	
	a) $f(a) + (x-a)\frac{f'(a)}{1!} + (x-a)^2\frac{f''(a)}{2!} + \cdots$	b) $f(a) - (x-a)\frac{f'(a)}{1!} - (x-a)^2\frac{f''(a)}{2!} - \cdots$
	c) $f(a) + (x+a)\frac{f'(a)}{1!} + (x+a)^2\frac{f''(a)}{2!} + \cdots$	d) $f(x) + (x+a)\frac{f'(x)}{1!} + (x+a)^2\frac{f''(x)}{2!} + \cdots$
13.	If the curvature of a function $f(x)$ is zero, then which of the following functions could be $f(x)$?	
	a) $ax + b$	b) $ax^2 + bx + c$
	c) $\sin x$	d) $\cos x$
14.	The angle between the radius vector and tangent for the vector $r = ae^{\theta cot\alpha}$ is :	
	a) $\tan\alpha$	b) $\cot\alpha$
	c) α	d) θ
15.	The angle between the radius vector and tangent for the vector $r = a\theta$ is	

	a) $\tan^{-1}\frac{1}{\theta}$	b) $\tan^{-1}\theta$
	c) r	d) $\frac{a}{\theta}$
16.	The radius of curvature to the curve $x = at^2, y = 2at$ when $t = 0$ is	
	a) $2a$	b) a
	c) 2	d) $\frac{a}{2}$
17.	The radius of curvature for the curve $x = f(t), y = g(t)$ is ρ	
	a) $\frac{(x'^2+y'^2)^{3/2}}{x'y''+y'x''}$	b) $\frac{(x'^2-y'^2)^{3/2}}{x'y''-y'x''}$
	c) $\frac{(x'^2+y'^2)^{2/3}}{x'y''+y'x''}$	d) $\frac{(x'^2+y'^2)^{3/2}}{x'y''-y'x''}$
18.	The curvature of the function $f(x) = x^3 - x + 1$ at $x = 1$ is _____	
	a) $\frac{6}{5}$	b) $\frac{6}{5}$
	c) $\frac{6}{5^{3/2}}$	d) $\frac{3}{5^{3/2}}$
19.	The radius of curvature for the curve $r = f(\theta)$ is ρ =_____	
	a) $\frac{(r^2+r_1^2)^{3/2}}{r^2+2r_1^2-rr_2}$	b) $\frac{(r^2+r_1^2)^{3/2}}{r^2-2r_1^2-rr_2}$
	c) $\frac{(r^2-r_1^2)^{3/2}}{r^2+2r_1^2-rr_2}$	d) $\frac{(r^2+r_1^2)^{2/3}}{r^2+2r_1^2-rr_2}$
20.	If the curvature of a curve increases then , the radius of curvature	
	a) increases	b) decreases
	c) constant	d) none of these

Unit 2 – Partial Differentiation		
1.	If $f(x,y,z) = x^2 + xyz + z$, then the value of f_x at (1,1,1)	
	a) 0	b) 1
	c) 3	d) -1
2.	If $z = 3x^2y + 5y$ then $\frac{\partial^2 z}{\partial x \partial y}$ is :	
	a) $3xy$	b) $6x$
	c) $3x + 5$	d) $6xy$
3.	The function $f(x,y) = x^2 + y^2 + 6x - 12$ has an extreme value at which one of the following intervals:	
	a) $(-3,0)$	b) $(0,3)$
	c) $(0,0)$	d) $(-3,-3)$

4	If $u = y^x$ then $\frac{\partial u}{\partial x}$ is	
	a) y^x	b) $y^x logy$
	c) $y^x logx$	d) xy^{x-1}
5	If $f(x,y,z) = x + y + z - log\ z\ ,\ then\ \ f_z(1,1,2)$	
	a) $1/2$	b) 1
	c) 2	d) 0
6	If $f(x,y,z) = cos\ (xy)\ +\ sin\ (y)\ + z,$ then $f_x\left(\frac{1}{2}, \frac{\pi}{2}, 7\right)$	
	a) $-\frac{\pi}{2\sqrt{2}}$	b) $\frac{\pi}{4\sqrt{3}}$
	c) $-\frac{\pi}{4}$	d) $\frac{\pi}{\sqrt{2}}$
7	If $x = rcos\theta, y = rsin\theta$ then the value of $\frac{\partial(x,y)}{\partial(r,\theta)}$ is :	
	a) r	a) 1
	c) 0	c) r^2
8	If $u = y^x$ then $\frac{\partial u}{\partial y}$ is :	
	a) y^x	b) $y^x\ log\ log\ y$
	c) $y^x\ log\ log\ x$	d) xy^{x-1}
9	If $u = e^x cos\ y$ and $v = e^x sin\ y$ the value of $J\left(\frac{u,v}{x,y}\right)$ is :	
	a) e^{2x}	b) $\frac{e^{2x}}{2}$
	c) $\frac{-e^{2x}}{2}$	d) e^{-2x}
10	If $u = x^3 + y^3$, then $\frac{\partial^2 u}{\partial x \partial y}$ is :	
	a) -3	b) 3
	c) 0	d) $3x + 3y$
11	If $f(x,y,z) = e^{xyz}$ then $f_x(2,2,2)$ is:	
	a) $4e^8$	b) $2e^8$
	c) $8e^8$	d) e^8
12	If $x = uv, y = \frac{u}{v}$ then $\frac{\partial(x,y)}{\partial(u,v)}$ is :	
	a) $-2\frac{u}{v}$	b) $-2\frac{v}{u}$
	c) 0	d) 1
13	If $f(x,y,z) = x^2 + xyz + z$ then f_x at (1,0,1)	

	a) 1	b) 2
	c) 3	d) 0

14	If $z = log(x^2 + y^2)$ then $z_y(1,1)$ is	
	a) 1	b) 2
	c) 0	d) 3
15	If $J_1 = \frac{\partial(u,v)}{\partial(x,y)}$, $J_2 = \frac{\partial(x,y)}{\partial(u,v)}$ then $J_1 J_2$ is	
	a) 2	b) 0
	c) 1	d) -1
16	If $f(x,y) = sin(x) + cos(y) + xy^2$ $where$ $x = cos(t)$; $y = sin(t)$, then $\frac{df}{dt}$ at $t = \pi / 2$ is :	
	a) 2	b) -2
	c) 1	d) 0
17	The Jacobian of p, q, r w.r.t x, y, z given $p = x + y + z$, $q = y + z$, $r = z$ is	
	a) 0	b) 1
	c) 2	d) -1
18	If $f(x,y,z) = x^2 + y^2 + z^2$ then $f_{xx}(2,4,2)$ is :	
	a) 1	b) 2
	c) 4	d) 8
19	Given $u = x + y$, $v = xy$ then the Jacobian $\frac{\partial(u,v)}{\partial(x,y)}$ is :	
	a) $x + y$	b) 0
	c) $x - y$	d) 1
20	If $z = 6x^2y^2 + 8x^2$ then $\frac{\partial z}{\partial x}$ is :	
	a) $6y^2 + 16x$	b) $6x^2 + 8x^2$
	c) $12xy^2 + 16x$	d) $6x^2 + 6y^2 + 8x^2$
21	If $f(x,y) = 2x^3 - 4y^2$, what is the value of f_x and f_y at (3,2)	
	a) $f_x = 54$, $f_y = -16$	b) $f_x = 16$, $f_y = -54$
	c) $f_x = 54$, $f_y = 0$	d) $f_x = -54$ $f_y = -16$
22	If $u = x + y$ and $v = x - y$ the value of $J\left(\frac{u,v}{x,y}\right)$ is :	
	a) -2	b) 2

	c) 1	d) 0
23	If $f(x,y) = x^2y + x$ where $x = t$ and $y = t^2$ then the value of $\frac{df}{dt}$ when $t = 1$ is :	
	a) 2	b) 4
	c) 1	d) 5
24	If $f(x,y) = sin(xy + x^3y)$ then f_x at (0,1) is:	
	a) 2	b) 5
	c) 1	d) 0
25	If $z = 3xy + 4x^2$,then value of $\frac{\partial z}{\partial x}$ *is*	
	a) $3y + 8x$	b) $3x + 4x^2$
	c) $3xy + 8x$	d) $3y + 3x + 8x$
26	Given $f_{xx} = x^2,\ f_{xy} = 8,\ f_{yy} = y^2$, then the nature of the stationary point at $(\sqrt{2},\sqrt{2})$ is,	
	a) Maximum	b) Minimum
	c) Saddle point	d) No maxima & minima
27	If $f(x,y) = x + y$,*where* $x = \sin t$; $y = \cos t$ then the value of $\frac{df}{dt}$ at t = $\pi/2$	
	a) 2	b) -2
	c) 1	d) 0

	UNIT – III - Ordinary differential equation	
	-	
1.	The order and degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^2 + y = 0$ is _____ and _____.	
	a) 1 and 3	b) 2 and 3
	c) 2 and 2	d) 3 and 3
2.	The order and degree of the differential equation $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2} = \frac{d^2y}{dx^2}$ is _____ and _____.	
	a) 1 and 1	b) 2 and 1
	c) 2 and 2	d) 1 and 3
3.	The order and degree of the differential equation: $\frac{d^2y}{dx^2} = \frac{\left(\frac{dy}{dx}\right)^2 + 1}{x}$ are _____ and _____.	

	a) 2 and 1	b) 1 and 2
	c) 2 and 3	d) 2 and 2
4.	The differential equation having order 3 and degree 2 is _____.	
	a) $\left(\frac{d^3y}{dx^3}\right)^2 + x\left(\frac{dy}{dx}\right)^2 = 0$	b) $\frac{d^3y}{dx^3} + \frac{dy}{dx} + y = 0$
	c) $\frac{d^3y}{dx^3} + y = 0$	d) $\frac{d^3y}{dx^3} + \left(\frac{dy}{dx}\right)^2 = 0$
5.	The differential equation having order 2 and degree 1 is _____.	
	a) $\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^2 = 0$	b) $\frac{d^2y}{dx^2} + \sqrt{\frac{dy}{dx}} = 0$
	c) $\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = 0$	d) $\left(\frac{d^2y}{dx^2}\right)^2 + \frac{dy}{dx} = 0$
6.	The integrating factor (I.F.) for $\frac{dy}{dx} + \left(\frac{2}{x}\right)y = \sin x$ is	
	a) x^2	b) e^{2x}
	c) $\frac{1}{x^2}$	d) x^2
7.	The integrating factor for $\frac{dy}{dx} + y\tan x = x$ is	
	a) $\sec x$	b) $sec^2\, x$
	c) $\tan x$	d) $e^{x\tan x}$
8.	For Bernoulli DE $\frac{dy}{dx} + P(x)y = Q(x)y^n$, the linearizing variable is :	
	a) y^{1-n}	b) y^{n-1}
	c) y^n	d) y^{-n}
9.	Which equation is exact?	
	a) $(y\cos x)dx + (x\sin y)dy = 0$	b) $(2y)dx + (3x)dy = 0$
	c) $(x^2 - y)dx + (y^2)dy = 0$	d) $(e^x y)dx + (e^x)dy = 0$
10.	The differential equation $(2xy)\,dx + (x^2 + 1)\,dy = 0$ is exact because _____.	
	a) $\frac{\partial M}{\partial y} = 2x$ and $\frac{\partial N}{\partial x} = 2x$	b) $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$
	c) $\frac{\partial M}{\partial x} = 2y\ and\ \frac{\partial N}{\partial x} = 1$	d) Both M(x, y) and N(x, y) are differentiable functions of x and y
11.	Which of the following function is the solution of the differential equation $y' + y = 0$.	
	a) $y = e^{2x}$	b) $y = e^x$
	c) $y = e^{-x}$	d) $y = e^{-2x}$

12.	Which of the following function is not a solution of the differential equation $y'' + y = 0$?	
	a) $y = 15\cos x$	b) $y = 2\sin x$
	c) $y = \cos x$	d) $y = \sin 2x$
13.	The complementary function of the differential equation $(D-3)^2 y = e^{3x}$ is :	
	a) $(C_1 + C_2)xe^{3x}$	b) $(C_1 + C_2 x)e^{3x}$
	c) $(C_1 e^{3x} + C_2 e^{-3x})$	d) $(C_1 x + C_2 x^2)e^{3x}$
14.	The differential equation whose auxiliary equation has the roots 0, 1,-1 is :	
	a) $(D^3 + 2D^2 + D)y = 0$	b) $(D^3 - D^2 - D)y = 0$
	c) $(D^3 - 2D^2 - D)y = 0$	d) $(D^3 - D)y = 0$
15.	The differential equation whose auxiliary equation has the roots 1,-1 is :	
	a) $y'' - y = 0$	b) $y'' + y = 0$
	c) $y' + y = 0$	d) $y'' - y' - y = 0$
16.	The particular integral of $(D^2 + 3D - 4)y = 12e^{2x}$ is :	
	a) e^{2x}	b) $2e^{2x}$
	c) $3e^{2x}$	d) $\frac{e^{2x}}{2}$
17.	$y = C_1 e^{-6x} + C_2 e^{2x}$ is the general solution of the equation :	
	a) $y'' + 4y' - 12y = 0$	b) $y'' + 4y' + 12y = 0$
	c) $y'' - 4y' + 12y = 0$	d) $y'' - 4y' - 12y = 0$
18.	The integrating factor for the differential equation: $\frac{dy}{dx} + y \cdot \frac{1}{x} = \log x$ is	
	a) $\log x$	b) x
	c) $\frac{1}{x}$	d) x^2
19.	For the differential equation $y'' - y = e^x$, choose $y_1 = e^x$, $y_2 = e^{-x}$. Then the Wronskian W is	
	a) 1	b) 0
	c) -2	d) $e^x - e^{-x}$
20.	The solution of $y'' + y = 0$ is :	
	a) $y = C_1 e^x + C_2 e^{-x}$	b) $y = C_1 \cos x + C_2 \sin x$
	c) $y = C_1 x + C_2$	d) $y = C_1 e^{ix} + C_2 e^{-ix}$

21.	The partial differential equation of the expression $z = ax + by$, where a and b are arbitrary constants, is:	
	a) $z = px - qy$	b) $z = p + q$
	c) $z = pq$	d) $z = px + qy$
22.	Which of the following equations is a parabolic partial differential equation?	
	a) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$	b) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$
	c) $a^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$	d) $\frac{\partial^2 u}{\partial x^2} = 4\frac{\partial^2 u}{\partial t^2}$
23.	The partial differential equation of the expression $2z = ax^2 + by^2$ where a and b are arbitrary constants, is:	
	a) $2z = px + qy$	b) $z = p + q$
	c) $z = py + qx$	d) $z = px - qy$
24.	The following second order partial differential equation is: $2x\frac{\partial^2 u}{\partial x^2} + 3x\frac{\partial^2 u}{\partial y^2} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}$	
	a) parabolic	b) hyperbolic
	c) elliptic	d) circular
25.	The partial differential equation of the expression $z = f(x)$, where $f(x)$ is an arbitrary function, is:	
	a) $z = \frac{\partial z}{\partial x}$	b) $\frac{\partial z}{\partial y} = 0$
	c) $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial x} = 0$	d) $z = \frac{\partial z}{\partial y}$
26.	The following second order partial differential equation is: $x\frac{\partial^2 u}{\partial x^2} - x^2\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$	
	a) parabolic	b) hyperbolic
	c) elliptic	d) circular
27.	The partial differential equation of the expression $z = f(x + y)$, where $f(x + y)$ is an arbitrary function, is:	
	a) $p - q = 0$	b) $p + q = 0$
	c) $z = p + q$	d) $z = p - q$
28.	The following second order partial differential equation is: $xy\frac{\partial^2 u}{\partial x \partial y} - x^2\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = \cos(2x + 3y)$	
	a) parabolic	b) hyperbolic
	c) elliptic	d) circular

29.	The partial differential equation of the expression $z = ae^x - be^y$, where a and b are arbitrary constants, is:	
	a) $z = p + q$	b) $z = p - q$
	c) $p + q = 0$	d) $p - q = 0$
30.	Which of the following equations is a hyperbolic partial differential equation?	
	a) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$	b) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$
	c) $a^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$	d) $\frac{\partial^2 u}{\partial x^2} = 4\frac{\partial^2 u}{\partial t^2}$
31.	The partial differential equation of the expression $z = ax + be^y$, where a and b are arbitrary constants, is:	
	a) $z = p + q$	b) $z = px + q$
	c) $z = p + qy$	d) $z = p - q$
32.	Which of the following partial differential equations is not an elliptic partial differential equation?	
	a) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$	b) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$
	c) $a^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$	d) $x^2 \frac{\partial^2 u}{\partial x^2} + 4y^2 \frac{\partial^2 u}{\partial t^2} = 4xy$
33.	The partial differential equation of the expression $z = xe^a - by$, where a and b are arbitrary constants, is:	
	a) $z = px - qy$	b) $z = px + qy$
	c) $z = p + q$	d) $px + qy = 0$
34.	The following second order partial differential equation is: $y^2 \frac{\partial^2 u}{\partial x^2} - 2xy \frac{\partial^2 u}{\partial x \partial y} + x^2 \frac{\partial^2 u}{\partial y^2} - x^2 \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$	
	a) parabolic	b) hyperbolic
	c) elliptic	d) circular
35.	The partial differential equation of the expression $z = e^y g(x)$, where $g(x)$ an arbitrary function, is:	
	a) $z = \frac{\partial z}{\partial y}$	b) $z = \frac{\partial z}{\partial x}$
	c) $\frac{\partial z}{\partial x} = 0$	d) $\frac{\partial z}{\partial y} = 0$
36.	Which of the following equation is not a hyperbolic partial differential equation?	

	a) $-2\frac{\partial^2 u}{\partial x^2} - x^2\frac{\partial^2 u}{\partial y^2} = 0$	b) $\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = 0$
	c) $xy\frac{\partial^2 u}{\partial x^2} - 3xy\frac{\partial^2 u}{\partial y^2} = 0$	d) $\frac{\partial^2 u}{\partial x^2} - y^2\frac{\partial^2 u}{\partial y^2} = 0$
37.	The partial differential equation of the expression $2z = ax^2 + by$, where a and b are arbitrary constants, is:	
	a) $2z = px + 2qy$	b) $z = px + qy$
	c) $z = 2px + qy$	d) $z = px - qy$
38.	The following second order partial differential equation is: $y^2\frac{\partial^2 u}{\partial x^2} - 4xy\frac{\partial^2 u}{\partial x \partial y} - 4x^2\frac{\partial^2 u}{\partial y^2} - \sin^2 xy = 0$	
	a) parabolic	b) hyperbolic
	c) elliptic	d) circular
39.	The partial differential equation of the expression $z = ay + bx$, where a and b are arbitrary constants, is:	
	a) $z = px + qy$	b) $z = qx + py$
	c) $z = p + q$	d) $z = px - qy$
40.	Which of the following partial differential equation is an elliptic partial differential equation?	
	a) $-2\frac{\partial^2 u}{\partial x^2} - x^2\frac{\partial^2 u}{\partial y^2} = 0$	b) $\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = 0$
	c) $xy\frac{\partial^2 u}{\partial x^2} - 3xy\frac{\partial^2 u}{\partial y^2} = 0$	d) $\frac{\partial^2 u}{\partial x^2} - y^2\frac{\partial^2 u}{\partial y^2} = 0$
41.	f(x, y) = x^2 + xyz + z Find f_x at (1,1,1)	
	a)0	b)1
	c)3	d) -1
42.	f(x, y) = sin(xy) + x^2 ln(y) Find f_{yx} at (0, $\pi/2$)	
	a)33	b)0
	c)3	d)1
43.	The one-dimensional wave equation is:	
	a) $\partial u/\partial t = c^2\ \partial^2 u/\partial x^2$	b) $\partial^2 u/\partial t^2 = c^2\ \partial^2 u/\partial x^2$
	c) $\partial^2 u/\partial t^2 = \partial^2 u/\partial x^2$	d) $\partial u/\partial t = \partial u/\partial x$
44.	Which of the following is the standard form of the Laplace equation in 2D?	
	a) $\partial^2 u/\partial x^2 + \partial^2 u/\partial y^2 = 0$	b)$\partial u/\partial x + \partial u/\partial y = 0$

	c) $\partial u/\partial x = \partial u/\partial y$	d) $\partial^2 u/\partial x^2 - \partial^2 u/\partial y^2 = 0$
45.	Which type of PDE is the wave equation?	
	a) Parabolic	b) Elliptic
	c) Hyperbolic	d) None of the above
46.	The direct integration method is applicable when the PDE is:	
	a) Linear and homogeneous	b) Linear and first order
	c) Non-linear	d) Of second order with constant coefficients
47.	Direct integration method solves PDEs by:	
	a) Using Fourier series	b) Directly integrating partial derivatives step by step
	c) Transforming to ODEs	d) Using separation of variables

1.	**The value of $\int_0^1 \int_0^x dy\,dx$ is**			
	A)	1	B)	2
	C)	0	D)	$1/2$
2.	**The value of $\int_0^1 \int_0^{\sqrt{x}} dydx$ =_____**			
	A)	$\frac{2}{3}$	B)	$\frac{3}{2}$
	C)	$\frac{1}{3}$	D)	3
3.	**The value of $\int_0^{\frac{\pi}{2}} \int_1^2 rdrd\theta$ =_____**			
	A)	$\frac{\pi}{4}$	B)	π
	C)	$\frac{\pi}{2}$	D)	$\frac{3\pi}{4}$
4.	**Which of the following formula can be used to find area of a plane region in the cartesian form?**			
	A)	$A = \iint x\,dx\,dy$	B)	$A = \iint xy\,dx\,dy$
	C)	$A = \iint f(x,y)dx\,dy$	D)	$A = \iint dx\,dy$
5.	**The value of $\int_{-1}^1 \int_0^1 \int_{-1}^0 dx\,dy\,dz$ =**			
	A)	2	B)	0
	C)	-1	D)	1
6.	$\int_0^1 \int_0^{x^2} dy\,dx$ represents			

	A)	The area bounded by the parabola $y = x^2$, the line $x = 1$ and the $x - axis$.	B)	The area bounded by the parabolas $y = x^2$ and $y^2 = x$.
	C)	The area bounded by the parabola $y = x^2$, the line $y = 1$ and the $y - axis$.	D)	The area bounded by the parabola $y = x^2$, the line $y = 1$ and the $x - axis$
7.	$\int_0^1 \int_0^x e^y \, dy \, dx =$			
	A)	$e + 2$	B)	e
	C)	$e - 2$	D)	$2e$
8.	**The value of** $\int_0^c \int_0^b \int_0^a dz \, dy \, dx$ **is**			
	A)	$\frac{ab}{c}$	B)	$\frac{ab}{2c}$
	C)	abc	D)	ab
9.	**Area of a region *R* in terms of polar co-ordinate is**			
	A)	$A = \iint_R r \, dr \, d\theta$	B)	$A = \iint_R dr \, d\theta$
	C)	$A = \iint_R r^2 dr \, d\theta$	D)	$A = \iint_R (r + 1) \, dr \, d\theta$
10.	The value of $\int_0^1 \int_0^2 xy \;\; dx \, dy$ is			
	A)	2	B)	4
	C)	0	D)	1
11.	**After changing the order of integration, the integral value of** $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy \, dx$ **is same as**			
	A)	$\int_0^\infty \int_0^y \frac{e^{-y}}{y} dx \, dy$	B)	$\int_0^\infty \int_0^\infty \frac{e^{-y}}{y} dx \, dy$
	C)	$\int_0^x \int_o^y \frac{e^{-y}}{y} dx \, dy$	D)	$\int_0^1 \int_x^\infty \frac{e^{-y}}{y} dx \, dy$
12.	**In polar form, the integral** $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dxdy$ **is equal to**			
	A)	$\int_0^{\pi/2} \int_0^\infty e^{-r^2} \, dr \, d\theta$	B)	$\int_0^{\pi/2} \int_0^\infty e^{-r} \, dr \, d\theta$
	C)	$\int_0^{\pi/2} \int_0^\infty e^{-(r^2 + \theta^2)} \, dr \, d\theta$	D)	$\int_0^{\pi/2} \int_0^\infty e^{-r^2} \, r \, dr \, d\theta$
13.	**The value of** $\gamma\left(\frac{1}{2}\right) =$			
	A)	$\sqrt{\frac{\pi}{2}}$	B)	$\sqrt{\pi}$
	C)	$\frac{\sqrt{\pi}}{2}$	D)	$\frac{\pi}{2}$
14.	**Which of the following is true?**			
	A)	$\gamma(n + 1) = n\gamma(n)$	B)	$\gamma(n - 1) = n\gamma(n)$
	C)	$\gamma(n + 1) = n + 2\gamma(n)$	D)	$\gamma(n + 2) = n\gamma(n)$
15.	**The value of** $\int_0^{\frac{\pi}{2}} sin^3 \, x \, cos^4 \, x \, dx$ **is**			
	A)	$\frac{3\pi}{280}$	B)	$\frac{2}{3}$
	C)	$\frac{2}{35}$	D)	$\frac{35}{2}$
16.	The value of $\beta(3, 2)$ is:			

	A)	**1/12**	B)	**1/2**
	C)	**1/10**	D)	**1/24**
17.	For any $\boldsymbol{m > 0}$, $\boldsymbol{\beta(m,1) =}$			
	A)	1	B)	$\boldsymbol{m}$
	C)	0	D)	$\boldsymbol{\frac{1}{m}}$
18.	**If $\boldsymbol{m,n > 0}$ then $\frac{\beta(m,n+1)}{\beta(m+1,n)}$ equals**			
	A)	m/n	B)	n/m
	C)	1	D)	0
19.	**The value $\boldsymbol{\beta\left(\frac{1}{2},\frac{1}{2}\right)}$ is**			
	A)	1	B)	π
	C)	-1	D)	$\sqrt{\pi}$
20.	**The value of $\int_0^\infty e^{-x^2}\,dx$ is**			
	A)	$\sqrt{\pi}$	B)	$\sqrt{\frac{\pi}{2}}$
	C)	$\frac{\sqrt{\pi}}{2}$	D)	$\frac{\pi}{2}$